

US Patent & Trademark Office

Patent Public Search | Text View

United States Patent Application Publication

20250284192

Kind Code

A1

Publication Date

September 11, 2025

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PATTERN FORMING METHOD AND METHOD OF MANUFACTURING SEMICONDUCTOR DEVICE

Abstract

A pattern forming method includes: forming a first layer on an object, the first layer containing a first photocurable composition; forming a second layer on the first layer, the second layer containing a second photocurable composition different in composition from the first photocurable composition; pressing a first template having a surface with a pattern against the second layer, to bring the pattern into contact with a foreign matter inside or on a surface of the second layer, thereby moving at least part of the foreign matter into the first layer; and applying a first light to the first layer and the second layer through the first template to cure the first layer and the second layer.

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Appl. No.: 18/828164

Filed: September 09, 2024

Foreign Application Priority Data

JP	2024-037605	Mar. 11, 2024
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Publication Classification

Int. Cl.: G03F7/00 (20060101); B29C43/02 (20060101); B29C43/18 (20060101); B29C43/20 (20060101); B29L31/34 (20060101); H01L21/027 (20060101)

U.S. Cl.:

Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION

[0001] This application is based upon and claims the benefit of priority from Japanese Patent Application No. 2024-037605, filed on Mar. 11, 2024; the entire contents of which are incorporated herein by reference.

FIELD

[0002] Embodiments relate to a pattern forming method and a method of manufacturing a semiconductor device.

BACKGROUND

[0003] A known example method of manufacturing a semiconductor device uses a technique of forming a fine pattern using nanoimprint lithography (NIL).

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0004] FIG. 1 is a perspective view for explaining an example structure of a template.

[0005] FIG. 2 is a schematic cross-sectional view for explaining the example structure of the template.

[0006] FIG. 3 is a schematic plan view for explaining an example of an object.

[0007] FIG. 4 is a schematic cross-sectional view for explaining a first embodiment.

[0008] FIG. 5 is a schematic cross-sectional view for explaining the first embodiment.

[0009] FIG. 6 is a schematic cross-sectional view for explaining the first embodiment.

[0010] FIG. 7 is a schematic cross-sectional view for explaining the first embodiment.

[0011] FIG. 8 is a schematic cross-sectional view for explaining the first embodiment.

[0012] FIG. 9 is a schematic cross-sectional view for explaining the first embodiment.

[0013] FIG. 10 is a schematic cross-sectional view for explaining the first embodiment.

[0014] FIG. 11 is a schematic cross-sectional view for explaining the first embodiment.

[0015] FIG. 12 is a schematic cross-sectional view for explaining the first embodiment.

[0016] FIG. 13 is a schematic cross-sectional view for explaining the first embodiment.

[0017] FIG. 14 is a schematic cross-sectional view for explaining a further embodiment of the first embodiment.

[0018] FIG. 15 is a schematic cross-sectional view for explaining the further embodiment of the first embodiment.

[0019] FIG. 16 is a schematic cross-sectional view for explaining the further embodiment of the first embodiment.

[0020] FIG. 17 is a schematic cross-sectional view for explaining the further embodiment of the first embodiment.

[0021] FIG. 18 is a schematic cross-sectional view for explaining a second embodiment.

[0022] FIG. 19 is a schematic cross-sectional view for explaining the second embodiment.

[0023] FIG. 20 is a schematic cross-sectional view for explaining the second embodiment.

[0024] FIG. 21 is a schematic cross-sectional view for explaining the second embodiment.

DETAILED DESCRIPTION

[0025] A pattern forming method in an embodiment includes: forming a first layer on an object, the

first layer containing a first photocurable composition; forming a second layer on the first layer, the second layer containing a second photocurable composition different in composition from the first photocurable composition; pressing a first template having a surface with a pattern against the second layer, to bring the pattern into contact with a foreign matter inside or on a surface of the second layer, thereby moving at least part of the foreign matter into the first layer; and applying a first light to the first layer and the second layer through the first template to cure the first layer and the second layer.

[0026] Embodiments will be hereinafter explained with reference to the drawings. A relation between the thickness and planar dimension of each of components illustrated in the drawings, a thickness ratio among the components, and so on may be different from actual ones. Further, in the embodiments, substantially the same components are denoted by the same reference signs and an explanation thereof is omitted where appropriate.

[0027] A pattern formation method using NIL includes pressing a pattern of an original plate, which is called a template, onto an imprint material layer such as an ultraviolet curable resin on an object, and applying light such as ultraviolet light to the imprint material layer to cure it and thus transferring the pattern to the imprint material layer.

[0028] FIG. 1 is a perspective view for explaining an example structure of the template. FIG. 2 is a schematic cross-sectional view for explaining the example structure of the template. FIG. 1 and FIG. 2 illustrate a template 10. FIG. 1 and FIG. 2 illustrate an X-axis, a Y-axis orthogonal to the X-axis, and a Z-axis orthogonal to each of the X-axis and the Y-axis. FIG. 2 illustrates a part of a cross-section taken along a line segment A1-A2 illustrated in FIG. 1.

[0029] The template 10 includes a substrate 1 having a surface 1a and a surface 1b as illustrated in FIG. 1 and FIG. 2, the surface 1a having a mesa MS, and the surface 1b having a depression CO. The mesa MS is a portion where a device pattern is to be formed. Examples of the device pattern are not particularly limited and may include a line-and-space pattern and a pillar pattern. Example planar shapes of the mesa MS are not particularly limited and may include a rectangular shape. The surface 1a is a principal surface.

[0030] The substrate 1 contains, for example, quartz glass. The substrate 1 preferably transmits the aforementioned light.

[0031] FIG. 3 is a schematic plan view for explaining an example of the object. FIG. 3 illustrates an object 100. The object 100 has a surface 100a. The surface 100a is orthogonal to a Z-axis direction. A thickness direction of the object 100 is, for example, the Z-axis direction.

[0032] The object 100 is, for example, a stack formed by stacking films on a substrate such as a silicon wafer. The films include a conductive film or/and an insulating film. The stack is, for example, a device wafer (semiconductor substrate) in the middle of manufacturing a semiconductor device. Examples of the semiconductor device include a semiconductor memory device such as a NAND flash memory, an electrical device having a microstructure such as a microelectromechanical system (MEMS), and a magnetic recording medium. The configuration of the object 100 is not limited to the above configuration.

[0033] The surface 100a has a shot area 110. The shot area 110 is a unit area to which patterns are transferred all at once by pressing the template 10 against an imprint material layer in one imprint. FIG. 3 illustrates a plurality of the shot areas 110 on the surface 100a. The shot areas 110 are arrayed along an X-Y plane of the substrate. The shot areas 110 to be imprinted are selected, for example, in an order along arrows in FIG. 3 using an imprint apparatus for performing imprint. The surface 100a may have a space among the shot areas 110.

First Embodiment

[0034] Next, a first embodiment of a pattern forming method using a template and a method of manufacturing a semiconductor device will be explained. FIG. 4 to FIG. 13 are schematic cross-sectional views for explaining the first embodiment. FIG. 4 to FIG. 13 illustrate a part of an X-Z cross-section of the object 100 in the shot area 110.

[0035] In the first embodiment, as illustrated in FIG. 4, an organic layer **101** is formed above the object **100**. The organic layer **101** may be formed on a plurality of the shot areas **110** of the object **100**.

[0036] The organic layer **101** is an uncured layer containing an uncured material of a first photocurable composition. The uncured material also includes a semi-cured material which is not completely cured. Examples of the first photocurable composition include ultraviolet curable organic materials. Examples of the first photocurable composition include an spin on carbon (SOC) film. Examples of the organic materials include resist materials containing an acrylic group or a methacrylic group. A viscosity of the uncured material in the first photocurable composition is, for example, 100 mPa's or more and 1000000 mPa's or less at a temperature of 25° C.

[0037] The organic layer **101** can be formed by supplying a material containing the uncured material of the first photocurable composition to the surface of the object **100** using, for example, a spin coating method or an ink-jet method.

[0038] The viscosity of the organic layer **101** is preferably, for example, 100 mPa's or more and 1000 mPa's or less under the environment in forming the organic layer **101**. The organic layer **101** and the object **100** may be bonded by forming an adhesive layer before the formation of the organic layer **101** and forming the organic layer **101** on the adhesive layer. The adhesive layer may be, for example, a film containing carbon, and may contain, for example, a functional group such as a hydroxyl group (OH group), a carboxyl group (COOH group), or an amino group (NH₂ group).

[0039] Next, as illustrated in FIG. 5, a resist layer **102** is formed on the organic layer **101**. The resist layer **102** is an uncured layer containing an uncured material of a second photocurable composition different in composition from the first photocurable composition. Examples of the second photocurable composition include an ultraviolet curable resist material. The resist material may contain at least one element selected from the group consisting of silicon, aluminum, boron, phosphorus, sulfur, arsenic, and iron.

[0040] The resist layer **102** can be formed by supplying a material containing the uncured material of the second photocurable composition to the surface of the object **100** using, for example, the spin coating method or the ink-jet method.

[0041] The viscosity of the organic layer **101** is preferably higher than the viscosity of the resist layer **102** under the environment in forming the resist layer **102**. This can prevent mixture of the first and second photocurable compositions in an interface between the organic layer **101** and the resist layer **102**. The viscosity of the organic layer **101** may be adjusted by applying light having a light intensity, which is lower than a light intensity when applying light through a later-explained template, to the organic layer **101** before the formation of the resist layer **102** to increase the viscosity. The organic layer **101** at the time when the resist layer **102** is formed is an uncured layer which is not completely cured (or a semi-cured layer), and has viscous property.

[0042] FIG. 5 further illustrates a foreign matter **103**. Above at least one shot area **110** of the shot areas **110**, the foreign matter **103** may exist as illustrated in FIG. 5, for example, in or after the formation of the resist layer **102**. The foreign matter **103** is exposed, for example, inside or on the resist layer **102**. A part of the foreign matter **103** may be inside the organic layer **101**. Examples of the foreign matter **103** include a particle (foreign particle). The particle has a composition different from the composition of each of the organic layer **101** and the resist layer **102**. The particle contains, for example, at least one element selected from metal elements and carbon element.

[0043] A maximum particle size of the particle is, for example, 30 nm or more and less than 400 nm. A plurality of the foreign matters **103** may be exposed inside or on the resist layer **102**.

[0044] The thickness of the organic layer **101** is preferably larger than the maximum particle size of the particle. The thickness of the organic layer **101** is, for example, 100 nm or more and 400 nm or less.

[0045] The thickness of the resist layer **102** may be smaller than the thickness of the organic layer

101. The thickness of the resist layer **102** is, for example, 0.03 μm or more and 0.15 μm or less.

[0046] The thickness of the organic layer **101**, the thickness of the resist layer **102**, and the maximum particle size of the particle can be measured in a cross-sectional reflection electron image, the cross-sectional reflection electron image being obtained, for example, by observing a cross-section including a thickness direction (Z-axis direction) of a sample including the object **100**, the organic layer **101**, and the resist layer **102** using a scanning electron microscope (SEM).

[0047] In the case of forming the resist layer **102** using the ink-jet method, the resist layer **102** may be formed by discharging a droplet **102a** of a material containing the uncured material of the second photocurable composition. FIG. 6 illustrates a plurality of droplets **102a** on the organic layer **101**. In this event, the foreign matter **103** may be exposed, for example, to the surface of the organic layer **101**.

[0048] Next, as illustrated in FIG. 7, a template **10A** is arranged above the resist layer **102**. For the explanation of the template **10A**, the explanation of the template **10** can be used where appropriate. The template **10A** is arranged so that the surface **1a** is located facing downward. The movements of the template **10A** and the object **100** can be controlled using, for example, the imprint apparatus.

[0049] The template **10A** further has a protrusion **1c** on the surface of the mesa **MS** of the substrate **1**. The protrusion **1c** forms a device pattern **DP**. FIG. 7 illustrates a plurality of the protrusions **1c**, but the number of the protrusions **1c** is not limited to the number illustrated in FIG. 7. Further, the template **10A** may further have a plurality of depressions on the surface of the mesa **MS**.

[0050] Next, as illustrated in FIG. 8, the template **10A** is pressed against the resist layer **102** and the device pattern **DP** of the template **10A** is filled with the resist layer **102** to mold the resist layer **102**. The template **10A** may be pressed against the resist layer **102** so that a residual film **RLT** of the resist layer **102** is formed, for example, between the protrusion **1c** and the object **100**. In this case, part of the particle may be exposed from the surface of the organic layer **101** to remain inside the resist layer **102**. The maximum particle size of the particle is, for example, larger than the thickness of the residual film **RLT**. The thickness of the residual film **RLT** is, for example, 10 nm or more and 30 nm or less. The maximum particle size of the particle being the foreign matter **103** is, for example, larger than the thickness of the residual film **RLT** and smaller than the thickness of the organic layer **101**.

[0051] In the case where the foreign matter **103** exists, the protrusion **1c** comes into contact with the foreign matter **103**, but the organic layer **101** is the uncured layer and has a viscosity at such a degree that the foreign matter **103** can move, so that at least part of the foreign matter **103** moves into the organic layer **101**. The whole foreign matter **103** may be moved to into the organic layer **101** as illustrated in FIG. 8.

[0052] Next, as illustrated in FIG. 9, light **L1** is applied through the template **10A** with the template **10A** being pressed against the molded resist layer **102** to cure the organic layer **101** and the resist layer **102**, thereby fixing the foreign matter **103**. Examples of the light **L1** include ultraviolet light. The light **L1** has a wavelength capable of sensitizing the first photocurable composition and the second photocurable composition.

[0053] Next, as illustrated in FIG. 10, the template **10A** is separated from the resist layer **102**. This can form a cured resist layer **102** including a transferred pattern **TP** of the device pattern **DP** and a cured organic layer **101**. The viscosity of the organic layer **101** cured by the application of the light **L1** becomes higher than that before the application of the light **L1**, and is, for example, more than 1000 mPa's and 1000000 mPa's or less.

[0054] Next, as illustrated in FIG. 11, the organic layer **101** is processed using the cured resist layer **102**, and the object **100** is processed using the cured and processed organic layer **101**, whereby the transferred pattern **TP** including a depression **100b** is formed on the object **100** as illustrated in FIG. 12. This forms a desired pattern on the object **100**. The depression **100b** may have an inverted shape of the protrusion **1c**. In this event, in an area of the object **100** located under the foreign matter **103**, no pattern is formed. This area becomes an area (defective chip) which cannot be used

as a semiconductor device. The object **100** can be processed by partially removing the object **100** and the organic layer **101** by dry etching using the cured organic layer **101** and the cured resist layer **102**. Examples of the dry etching include reactive ion etching (RIE). The remainder of the organic layer **101** after the processing and the foreign matter **103** are removed thereafter.

[0055] The object **100** and the organic layer **101** may be processed in stages by a plurality of times of etching. In this event, the organic layer **101** and the resist layer **102** having different etching rates is preferable because even a complicated pattern having a plurality of depressions and/or protrusions having different heights can be easily formed. For example, the composition of the organic layer **101** is made different from the composition of the resist layer **102**, thereby making it possible to give etching rates different between the organic layer **101** and the resist layer **102**.

[0056] Next, a metal material is buried in the depression **100b** of the transferred pattern TP to form a conductor **200** as illustrated in FIG. **13**. The metal material contains, for example, at least one metal element selected from the group consisting of copper, aluminum, tungsten, and titanium. The conductor **200** has a function, for example, as wiring having a dual damascene structure.

[0057] By the above process, a semiconductor device can be manufactured.

[0058] As explained above, the first embodiment includes pressing the foreign matter **103** inside or on the resist layer **102** by the device pattern DP of the template **10A** to move at least part of the foreign matter **103** into the organic layer **101**.

[0059] If the organic layer **101** is cured, the resist layer **102** is formed on the cured organic layer **101**, and then the template **10A** is pressed against the resist layer **102**, the foreign matter **103** cannot be moved into the organic layer **101** and the foreign matter **103** comes into contact with the device pattern DP of the template **10A** to break the template **10A** in some cases. If the template **10A** is broken, it becomes difficult to form the predetermined device pattern DP thereafter. This needs to add an inspection step of confirming the presence or absence of the foreign matter **103** is provided before pressing the template **10A**, and if the foreign matter **103** exists, processing of not pressing the template **10A** against the resist layer **102** for the corresponding shot area **110**.

[0060] In contrast to the above, in the first embodiment, the organic layer **101** is the uncured layer and has a viscosity at such a degree that the foreign matter **103** can move when the template **10A** is pressed against the resist layer **102**, so that by bringing the device pattern DP of the template **10A** into contact with the foreign matter **103**, it is possible to mold the resist layer **102** and move at least part of the foreign matter **103** into the organic layer **101**. This can prevent the breakage of the template **10A**.

[0061] Further, in the first embodiment, any inspection step for confirming the presence or absence of the foreign matter **103** after forming the organic layer **101** and before pressing the template **10A** against the resist layer **102** can be made unnecessary even if the foreign matter **103** exists, so that it is possible to simplify the manufacturing process of the semiconductor device without including the inspection step.

Further Embodiment of First Embodiment

[0062] Next, a further embodiment of the first embodiment will be explained. FIG. **14** to FIG. **17** are schematic cross-sectional views for explaining the further embodiment of the first embodiment. FIG. **14** to FIG. **17** illustrate a part of an X-Z cross-section of the object **100**.

[0063] In the further embodiment of the first embodiment, the template **10A** is arranged so that the surface of the mesa MS faces downward, above the resist layer **102** as illustrated in FIG. **14** as in the first embodiment. The further embodiment of the first embodiment is different in shape of the device pattern of the template **10A** as compared with the first embodiment. In the following, differences from the first embodiment will be explained, and for the other portions, the explanation in the first embodiment can be used where appropriate.

[0064] The template **10A** has a depression **1d** on the surface **1a**. The depression **1d** is provided on the surface of the mesa MS and forms the device pattern DP. The device pattern DP including the depression **1d** may be an inverted pattern of the device pattern DP including the protrusion **1c**. FIG.

14 illustrates a plurality of the depressions **1d**, but the number of the depressions **1d** is not limited to the number illustrated in FIG. **14**. The template **10A** may further have a plurality of protrusions on the mesa **MS**. For the other explanation of the template **10A**, the explanation of the template **10A** in the first embodiment can be used where appropriate.

[0065] Next, the template **10A** is pressed against the organic layer **101** as in the first embodiment to mold the resist layer **102**, and the light **L1** is applied through the template **10A** while the template **10A** being pressed against the molded resist layer **102** as illustrated in FIG. **15** to cure the organic layer **101** and the resist layer **102**.

[0066] Thereafter, as in the first embodiment, the organic layer **101** is processed using the cured resist layer **102**, and the object **100** is processed using the cured and processed organic layer **101**, whereby the transferred pattern **TP** is formed on the object **100**.

[0067] As explained above, in the further embodiment of the first embodiment, the organic layer **101** is the uncured layer and has a viscosity at such a degree that the foreign matter **103** can move when the template **10A** is pressed against the resist layer **102** as in the first embodiment, so that by bringing the device pattern **DP** of the template **10A** into contact with the foreign matter **103**, it is possible to mold the resist layer **102** and move at least part of the foreign matter **103** into the organic layer **101**. This can prevent the breakage of the template **10A**.

[0068] Further, in the further embodiment of the first embodiment, any inspection step

[0069] for confirming the presence or absence of the foreign matter **103** after forming the organic layer **101** and before pressing the template **10A** against the resist layer **102** is made unnecessary even if the foreign matter **103** exists, so that it is possible to simplify the manufacturing process of the semiconductor device without including the inspection step.

Second Embodiment

[0070] Next, a second embodiment of a pattern forming method using a template and a method of manufacturing a semiconductor device will be explained. FIG. **16** to FIG. **21** are schematic cross-sectional views for explaining the second embodiment. FIG. **16** to FIG. **21** illustrate a part of an X-Z cross-section of the object **100**.

[0071] In the second embodiment, the organic layer **101** is formed on the object **100** as in the first embodiment. For the explanation of the object **100** and the organic layer **101**, the explanation of the object **100** and the organic layer **101** in the first embodiment can be used where appropriate.

[0072] Above at least one shot area **110** of the shot areas **110**, the foreign matter **103** may exist in some cases as illustrated in FIG. **16** or FIG. **17** in or after the formation of the organic layer **101**. The foreign matter **103** is exposed, for example, inside or on the organic layer **101**. For the other explanation of the foreign matter **103**, the explanation of the foreign matter **103** in the first embodiment can be used where appropriate.

[0073] FIG. **16** and FIG. **17** further illustrate a protrusion **104** formed due to the foreign matter **103**. The protrusion **104** is formed on the surface of the organic layer **101**. The protrusion **104** may be formed including part of the organic layer **101** because part of the organic layer **101** protrudes due to the foreign matter **103**, for example, as illustrated in FIG. **16**, or may be formed including part of the foreign matter **103** because of the foreign matter **103** protrudes from the surface of the organic layer **101** as illustrated in FIG. **17**.

[0074] Next, a template **10B** is arranged above the organic layer **101**, and then the template **10B** is pressed against the organic layer **101** to mold the organic layer **101** as illustrated in FIG. **18**.

[0075] For the explanation of the template **10B**, the explanation of the template **10** can be used where appropriate. The template **10B** is arranged so that the surface of the mesa **MS** is located facing downward. The movements of the template **10B** and the object **100** can be controlled using, for example, the imprint apparatus.

[0076] The template **10B** has no pattern on the mesa **MS**. The surface **1a** includes the surface of the mesa **MS**. The surface of the mesa **MS** of the template **10B** is preferably flat.

[0077] By pressing the template **10B** against the resist layer **102**, the protrusion **104** is pressed by

the surface of the mesaMS of the template **10B**. This can eliminate the protrusion **104** to make the surface (upper surface) of the organic layer **101** flat. In this event, the foreign matter **103** may be moved into the organic layer **101**. The whole foreign matter **103** may be moved into the organic layer **101**.

[0078] Next, as illustrated in FIG. **19**, light L2 is applied through the template **10B** with the template **10B** being pressed against the molded organic layer **101** to cure the organic layer **101**. This can fix the foreign matter **103** inside the organic layer **101**. Examples of the light L2 include ultraviolet light. The light L2 has a wavelength capable of sensitizing the first photocurable composition.

[0079] Next, as illustrated in FIG. **20**, the template **10B** is separated from the organic layer **101**. This can form a flattened and cured organic layer **101**. The surface roughness can be measured from a cross-sectional reflection electron image obtained, for example, by observing a cross-section including a thickness direction (Z-axis direction) of a sample including the object **100** and the organic layer **101** using the SEM.

[0080] Next, the resist layer **102** is formed on the flattened and cured organic layer **101** as illustrated in FIG. **21**. For the other explanation of the resist layer **102**, the explanation of the resist layer **102** in the first embodiment can be used where appropriate.

[0081] Next, the template **10A** is pressed against the resist layer **102** to mold the resist layer **102** by the same method as in the first embodiment, and the light L1 is applied through the template **10A** with the template **10A** being pressed against the molded resist layer **102** to cure the resist layer **102**. The device pattern DP of the template **10A** may include the depression **1d** as in the further embodiment of the first embodiment.

[0082] Thereafter, by the same method as in the first embodiment, the organic layer **101** is processed using the cured resist layer **102**, and the object **100** is processed using the cured organic layer **101**, whereby the transferred pattern TP of the device pattern DP is formed on the object **100**. The object **100** can be processed by partially removing the object **100**, the organic layer **101**, and the resist layer **102** by dry etching such as RIE using the cured organic layer **101** and the cured resist layer **102**. The remainder of the organic layer **101** after the processing and the foreign matter **103** are removed thereafter.

[0083] Next, a metal material is buried in the depression **100b** by the same method as in the first embodiment to form the conductor **200**.

[0084] By the above process, a semiconductor device can be manufactured.

[0085] As explained above, the second embodiment includes pressing the protrusion **104** formed on the organic layer **101** due to the foreign matter **103** inside or on the organic layer **101** by the template **10B** having the flat surface with no pattern to flatten the upper surface of the organic layer **101**, forms the resist layer **102** on the flattened organic layer **101**, and then performs imprint using the template **10A** having a pattern.

[0086] If the resist layer **102** is formed on the organic layer **101** with the protrusion **104** remaining and the template **10A** is then pressed against the resist layer **102**, the protrusion **104** comes into contact with the template **10A** after the curing of the organic layer **101** to break the template **10A** in some cases. If the template **10A** is broken, it becomes difficult to form the desired device pattern DP thereafter. Therefore, an inspection step of confirming the presence or absence of the foreign matter **103** is provided before pressing the template **10A**, and if the foreign matter **103** exists, processing of not pressing the template **10A** against the resist layer **102** for the corresponding shot area **110** is necessary.

[0087] In contrast to the above, in the second embodiment, the surface of the organic layer **101** is flattened before the template **10A** is pressed against the resist layer **102**, thereby making it possible to prevent the breakage of the template **10A** even if the template **10A** is pressed against the resist layer **102** after the curing of the organic layer **101**.

[0088] Further, in the second embodiment, any inspection step for confirming the presence or

absence of the foreign matter **103** after forming the organic layer **101** and before pressing the template **10A** against the resist layer **102** can be made unnecessary even if the foreign matter **103** exists, so that it is possible to simplify the manufacturing process of the semiconductor device without including the inspection step.

[0089] The second embodiment can be appropriately combined with the first embodiment.

[0090] While certain embodiments of the present invention have been described above, these embodiments have been presented by way of example only, and are not intended to limit the scope of the invention. Indeed, the novel embodiments described herein may be embodied in a variety of other forms; furthermore, various omissions, substitutions and changes in the form of the embodiments described herein may be made without departing from the spirit of the inventions. The accompanying claims and their equivalents are intended to cover such forms or modifications as would fall within the scope and spirit of the inventions.

Claims

1. A pattern forming method, comprising: forming a first layer on an object, the first layer containing a first photocurable composition; forming a second layer on the first layer, the second layer containing a second photocurable composition different in composition from the first photocurable composition; pressing a first template having a surface with a pattern against the second layer to bring the pattern into contact with a foreign matter inside or on a surface of the second layer, thereby moving at least part of the foreign matter into the first layer; and applying a first light to the first layer and the second layer through the first template to cure the first layer and the second layer.
2. The method according to claim 1, wherein the first layer is formed by supplying the first photocurable composition onto the object using a spin coating method or an ink-jet method.
3. The method according to claim 1, wherein the second layer is formed by supplying the second photocurable composition onto the object using a spin coating method or an ink-jet method.
4. The method according to claim 1, wherein the first photocurable composition contains an ultraviolet curable organic material.
5. The method according to claim 1, wherein the second photocurable composition contains at least one element selected from the group consisting of silicon, aluminum, boron, phosphorus, sulfur, arsenic, and iron.
6. The method according to claim 1, wherein: the foreign matter is a particle; and a thickness of the first layer is larger than a maximum particle size of the particle.
7. The method according to claim 1, wherein a viscosity of the first layer is higher than a viscosity of the second layer in forming the second layer.
8. A pattern forming method comprising: forming a first layer on an object, the first layer containing an uncured material of a first photocurable composition; pressing a first template having a flat surface against the first layer to flatten an upper surface of the first layer, the protrusion being caused by a foreign matter inside or on a surface of the first layer; applying a first light to the first layer through the first template to cure the first layer; forming a second layer on the cured first layer, the second layer containing an uncured material of a second photocurable composition different in composition from the first photocurable composition; pressing a second template having a surface with a pattern against the second layer to bring the pattern into contact with the second layer; and applying a second light to the second layer through the second template to cure the second layer.
9. The method according to claim 8, wherein the first layer is formed by supplying the first photocurable composition onto the object using a spin coating method or an ink-jet method.
10. The method according to claim 8, wherein the second layer is formed by supplying the second photocurable composition onto the object using a spin coating method or an ink-jet method.

- 11.** The method according to claim 8, wherein the first photocurable composition contains an ultraviolet curable organic material.
- 12.** The method according to claim 8, wherein the second photocurable composition contains at least one element selected from the group consisting of silicon, aluminum, boron, phosphorus, sulfur, arsenic, and iron.
- 13.** The method according to claim 8, wherein: the foreign matter is a particle; and a thickness of the first layer is larger than a maximum particle size of the particle.
- 14.** The method according to claim 8, wherein a viscosity of the first layer is higher than a viscosity of the second layer in forming the second layer.
- 15.** A method of manufacturing a semiconductor device, comprising: forming a first cured layer and a second cured layer on a substrate by a first pattern forming method or a second pattern forming method; and etching the substrate using the first cured layer and the second cured layer to partially remove the substrate, the first pattern forming method comprising: forming a first layer on the substrate, the first layer containing a first photocurable composition; forming a second layer on the first layer, the second layer containing a second photocurable composition different in composition from the first photocurable composition; pressing a first template having a surface with a pattern against the second layer to bring the pattern into contact with a foreign matter inside or on a surface of the second layer, thereby moving at least part of the foreign matter into the first layer; and applying a first light to the first layer and the second layer through the first template to cure the first layer and the second layer, and the second pattern forming method comprising: forming a third layer on the substrate, the third layer containing an uncured material of the first photocurable composition; pressing a second template having a flat surface against the third layer to flatten an upper surface of the third layer, the protrusion being caused by the foreign matter; applying a second light to the third layer through the second template to cure the third layer; forming the fourth layer on the cured third layer; pressing the first template against the fourth layer to bring the pattern into contact with the fourth layer; and applying the first light to the fourth layer through the first template to cure the fourth layer.
- 16.** The method according to claim 15, wherein the first layer or the third layer is formed by supplying the first photocurable composition onto the substrate using a spin coating method or an ink-jet method, and the second layer or the fourth layer is formed by supplying the second photocurable composition onto the substrate using a spin coating method or an ink-jet method.
- 17.** The method according to claim 15, wherein the first photocurable composition contains an ultraviolet curable organic material.
- 18.** The method according to claim 15, wherein the second photocurable composition contains at least one element selected from the group consisting of silicon, aluminum, boron, phosphorus, sulfur, arsenic, and iron.
- 19.** The method according to claim 15, wherein: the foreign matter is a particle; and a thickness of the first layer or the third layer is larger than a maximum particle size of the particle.
- 20.** The method according to claim 15, wherein a viscosity of the first layer is higher than a viscosity of the second layer in forming the second layer, and a viscosity of the third layer is higher than a viscosity of the fourth layer in forming the fourth layer.
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